ResumeX

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Table of Content

1. Problem Statement
2. Objectives
3. Scope
4. Literature Review
5. UML Diagram
6. Methodology
7. Tools and Technology
8. Projected Output

Problem Statement

Semicolon Company

Semicolon Company

Semicolon Company

Semicolon Company

Solution

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Resume Ranking System

Objective

- Faster, Fairer Hiring
- Criteria-Based Resume Evaluation
- Bias-Free Resume Ranking
- Optimized Candidate Matching
- Efficient Resume Filtering

Scope

- Secure Registration, Role-Based Access
- Resume Analysis, Ranking, Scoring
- User-Friendly Interface, Admin Dashboard
- Django Backend for Database and Resume Processing

Methodology

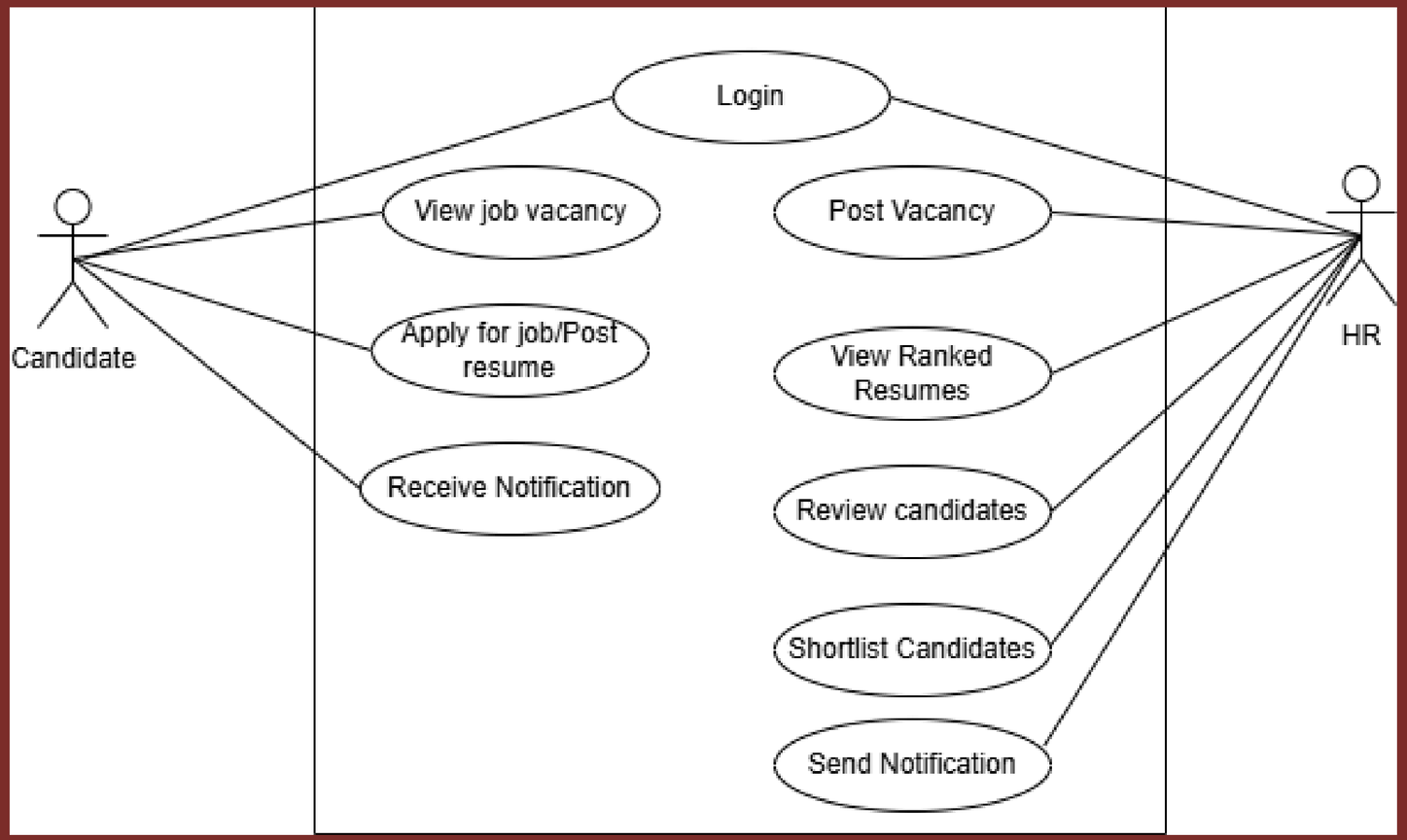

Fig 1: UseCase Diagram

Methodology

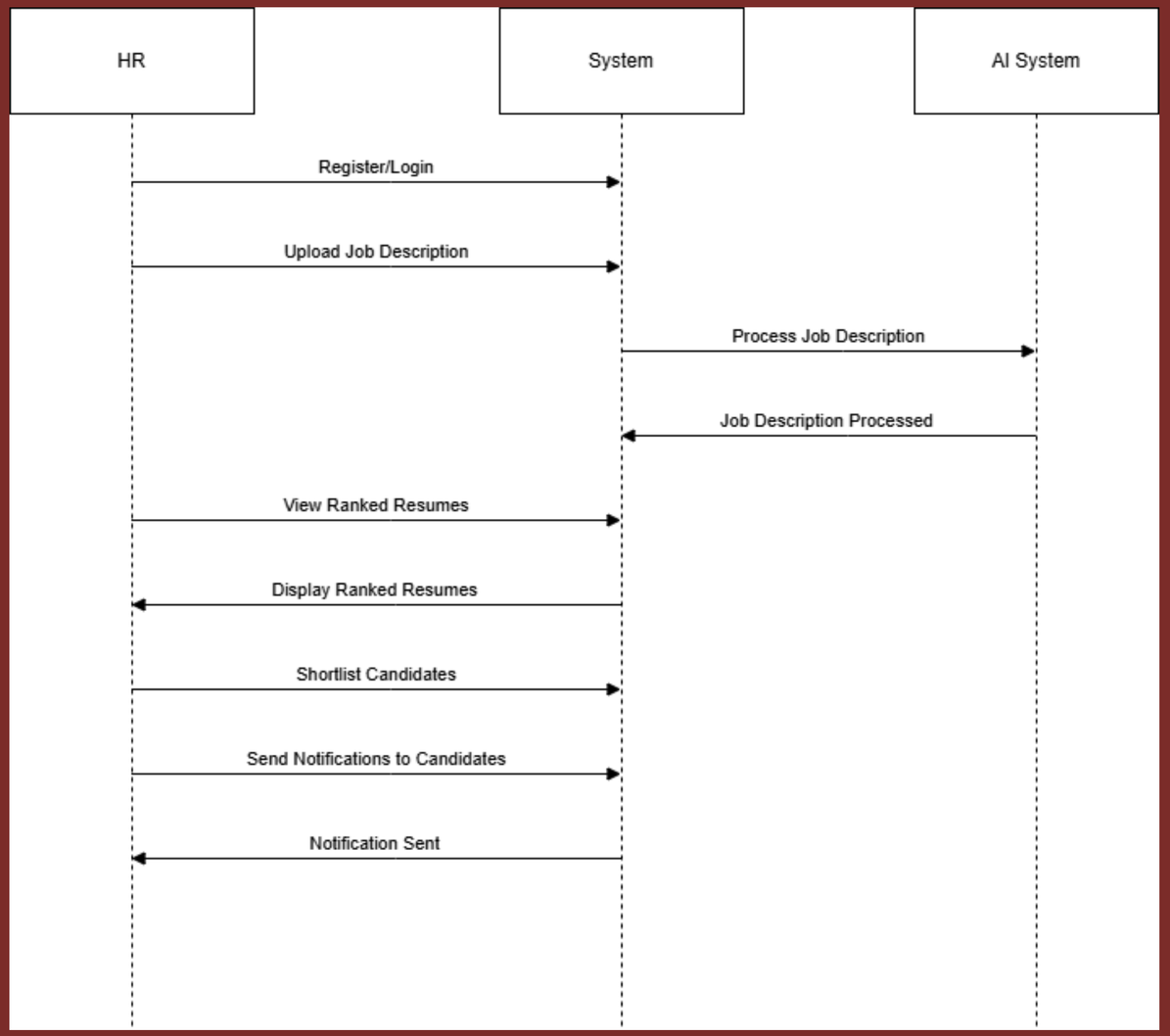

Fig 2: Sequence Diagram of HR

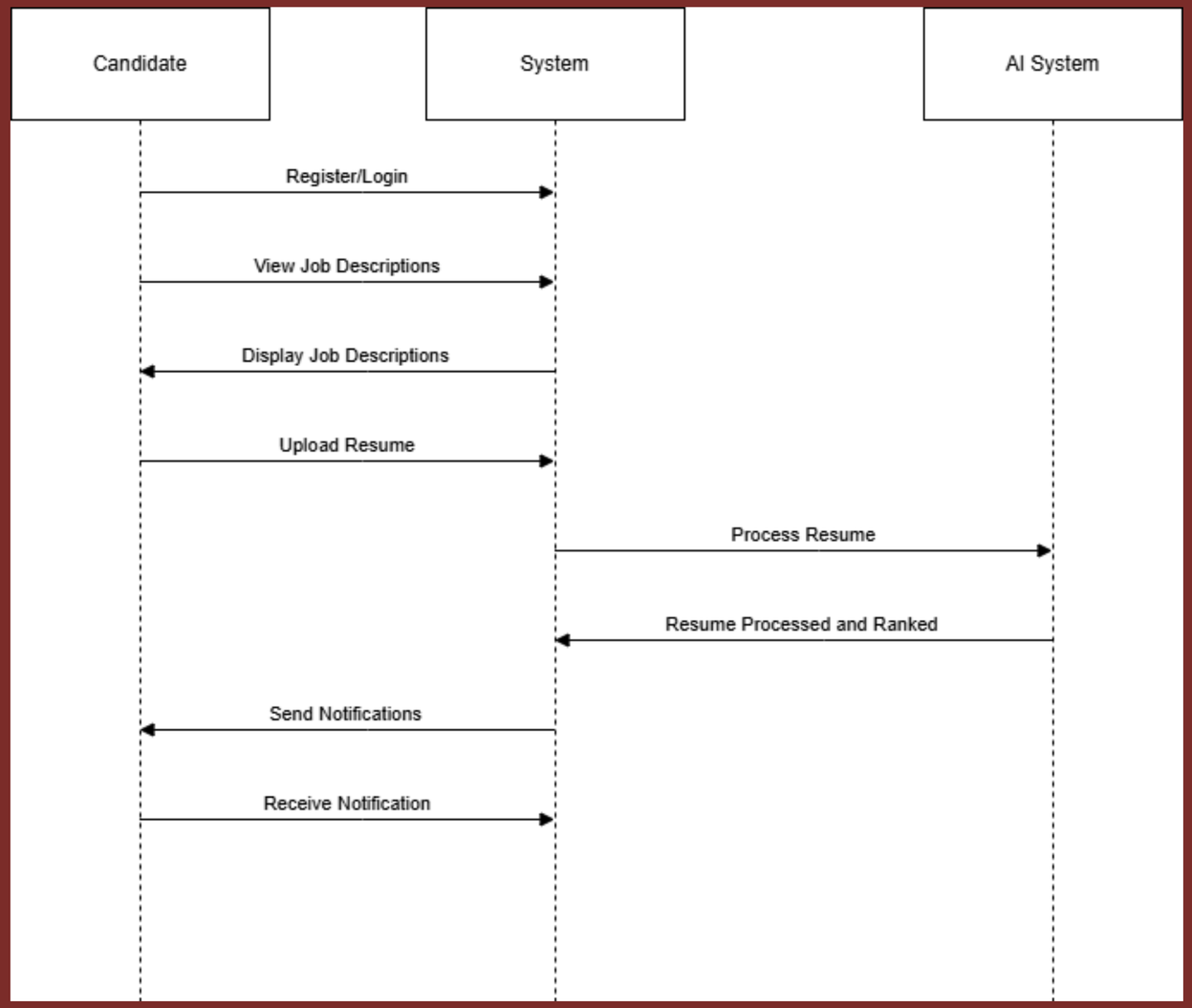

Fig 3: Sequence Diagram of Candidate

Literature Review

Related Companies:

<u>Textkernel</u>

What it does:
Textkernel uses AI to read and organize data from resumes and match it with job descriptions. It uses powerful models like GPT-3.5 and BERT for accurate data extraction and matching.

Difference from our project:
Textkernel focuses on parsing and matching resumes to jobs, while our system is specifically about ranking resumes based on predefined criteria. Textkernel uses advanced, pre-trained models that require more resources.

Literature Review

Related Companies:

<u>Hiretual</u>

What it does:

Hiretual is an AI recruitment platform that helps companies find candidates by searching profiles across multiple platforms. It uses semantic search to make better recommendations.

<u>Difference from our project:</u>

Hiretual focuses on finding and engaging candidates from different platforms, while our system only ranks resumes submitted for a specific job..

Literature Review

RELATED PROJECT:

Resume Ranking System (Galey et al.):

What it does:
This project uses NLP to rank resumes based on the skills and experience mentioned in the resume. It uses libraries like NLTK and SpaCy for natural language processing and MERN for web development.

Difference from our project:
This system ranks resumes like ours but uses different tech stacks and focuses on creating a web platform.

Algorithms

- NLP

Prepares and processes textual data from resumes and job descriptions by tokenizing, removing stopwords, applying TF-IDF/embeddings, and converting text into feature vectors for comparison.

- **Cosine Similarity**

Measures the cosine of the angle between the job description and resume vectors.

$$cos(\theta) = \frac{A \cdot B}{||A||\ ||B||}$$

Algorithms

- KNN

Finds the most similar resumes to a given job description by selecting the top K closest resumes using distance metrics (e.g., Euclidean).

Steps to Adapt KNN for Ranking:

- Calculate Distances

- Sort Neighbors:
  The smallest distance (or highest similarity score) indicates a better match.

- Rank Resumes:
  Assign ranks based on their distances or scores. The closest resumes are ranked higher.

Tools and Technology

Tools and Technology

1. Frontend -> React.js

2. Backend -> Django

3. Deployment and Devlopment-> Github

4. Resume Ranking -> AI/ML

5. Communitcation-> Discord

Thank You!